

Analytics Hub enables data sharing

So far, you've learned that a data library is a collection of datasets that are stored and shared both internally within your own organization, and externally to outside users. Services like Google's Analytics Hub allow you to share datasets, machine learning models, and other analytics assets, which can then be combined with your own internal data. In this reading, you'll learn more about data libraries, and how Analytics Hub can support your work as a cloud data analyst.

How data libraries work

Using a service like Analytics Hub enables you to publish datasets and make them available to both teams within your organization, and external stakeholders. You can publish both tables and saved views in BigQuery to make them accessible to others. To do this, a data manager curates datasets called exchanges, and makes them available for sharing. Then, these exchanges are organized by metadata to make searching for and finding data easier. The organization that published the data can view those who subscribe to their exchanges.

Benefits of data libraries

Whether you are a publisher or end-user, there are several benefits to using Analytics Hub. Review the chart to learn more about benefits for publishers and stakeholders:

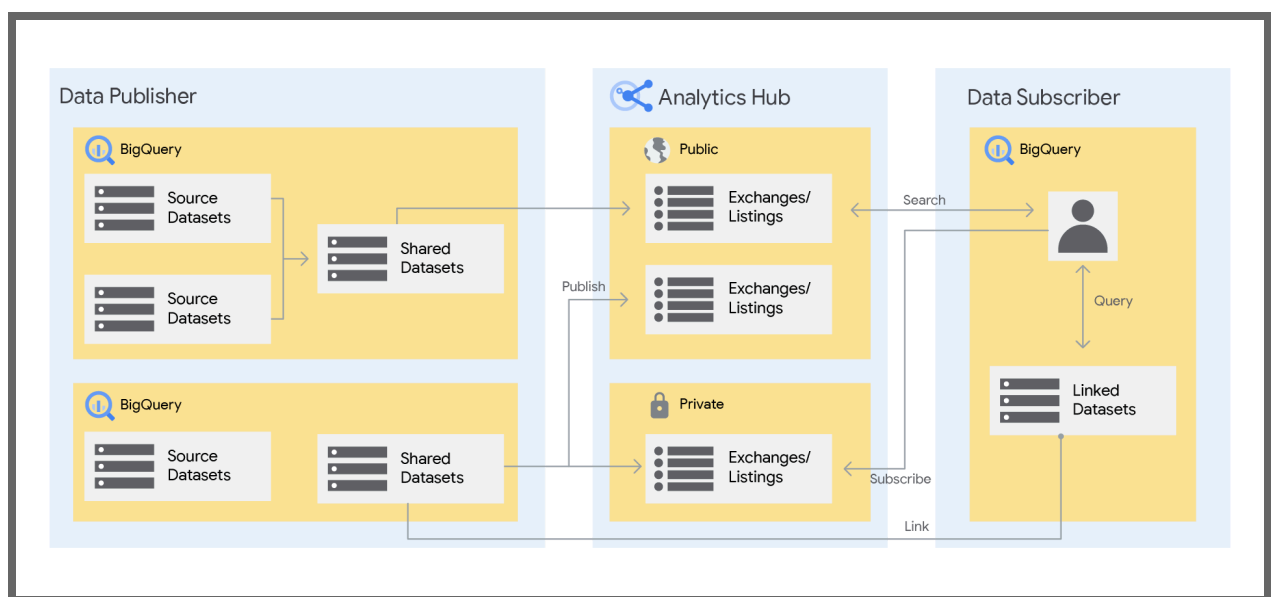
	Publishers	Subscribers
Benefit	• Can list datasets for sharing without having to replicate the entire dataset, saving on storage needs • Can create data exchanges and give permissions to subscribers to access datasets and accompanying analytics	• Can search for data, combine datasets with existing datasets, and use BigQuery to organize data • Can browse through datasets and request shared access from the publishers

Analytics Hub architecture

The basic premise of Analytics Hub architecture for a data library is the separation of the compute and storage resources available in BigQuery. This means that organizations who publish the data can share it with as many external users as they want, without having to

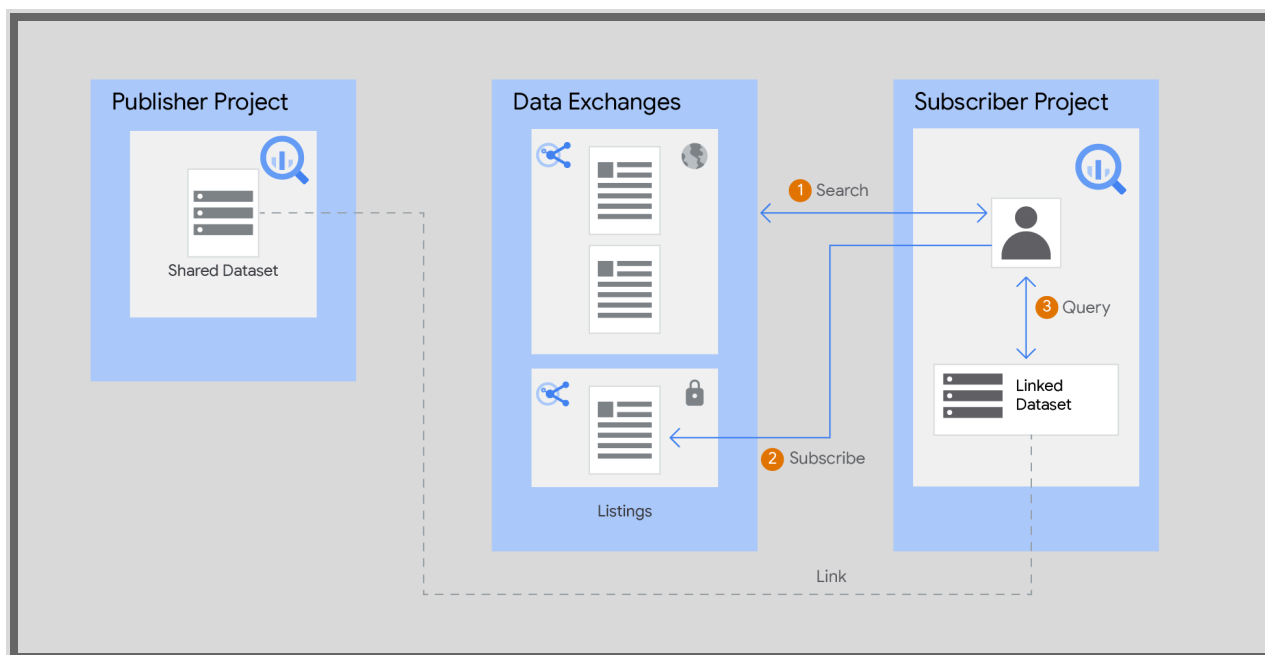
duplicate and store the data for each user. But, subscribers don't have to use their own databases to store the data, and only use their own compute services based on the queries they run.

Consider this example: A worldwide sports organization publishes some datasets for external use by journalists and coaches. Data is made available by the data publisher to the subscriber. The sports organization takes varied data and combines them into new datasets. Then, these datasets are listed in Analytics Hub as either public or private in exchanges for the journalists and coaches to use. All public data is available for external users to access via subscription. Data subscribers are then able to search these Exchanges for the data they need, and query the linked datasets.



Subscriber access

The Analytics Hub architecture makes it clear how the end user or subscriber accesses the data. While the user searches Analytics Hub exchanges to find the data they need, they query the data right from the publisher's dataset. This creates a linked dataset that the subscriber can access in their own projects without needing to physically copy the data from the publisher.



Key takeaways

Data libraries allow you to share data internally and externally without duplicating datasets. Organizations can group data into datasets and publish them as Exchanges to allow external users to subscribe to and access these published datasets. The datasets are hosted by the publishing organization, and external users can use their own compute resources to query the data. A tool like Google's Analytics Hub allows organizations to provide access to datasets from BigQuery.

Resources for more information

To learn more about how Analytics Hub allows you to publish and access exchanges, check out these links:

- Google Cloud documentation provides an Introduction to Analytics Hub, including an overview of how data is shared and accessed:
<https://cloud.google.com/bigquery/docs/analytics-hub-introduction#>